Linear Regression

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October 13, 2026

Outline

The Linear Model
- Simple linear regression
- History
- Multiple linear regression

Optimisation algorithms
- Gradient Descent
- Least-Squares

Regression libraries in Python
- sklearn
- statsmodels

Problems
- Interpretation Problem parameters
- Exercises

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Simple linear regression

Input and output

- Data pairs (x^t, y^t), $t = 1, \ldots, T$.
- Input $x^t \in \mathbb{R}$
- Output $y^t \in \mathbb{R}$.

Modelling the conditional expectation $\mathbb{E}[y^t|x^t]$

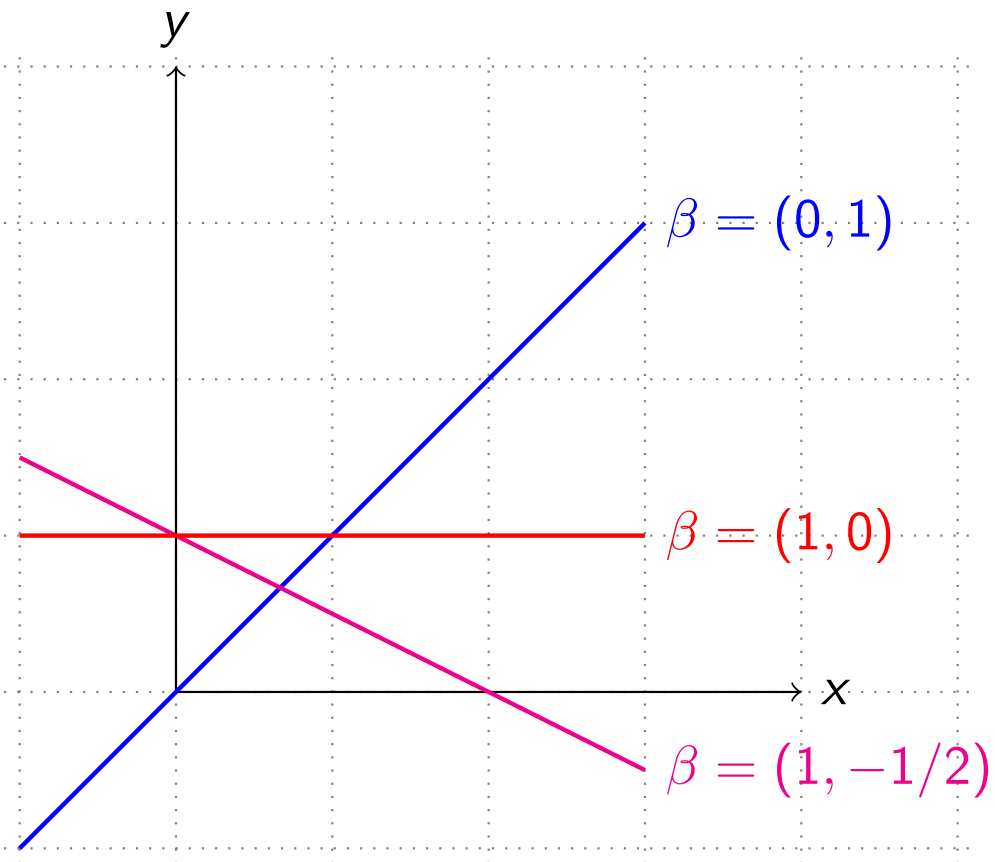
- Parameters $\beta_0, \beta_1 \in \mathbb{R}$
- Function $\pi_\beta : \mathbb{R} \to \mathbb{R}$, defined as

$$\pi_\beta(x^t) = \beta_0 + \beta_1 x_t$$

Probabilistic predictions: Modelling the conditional probability $\mathbb{P}[y^t|x^t]$

- $y^t = \mathbb{E}[y^t|x^t] + \epsilon_t$, with $\epsilon_t \in \mathbb{R}$ being zero-mean noise.
- Simplest model: variance $\sigma = \mathbb{V}(\epsilon) = \mathbb{E}[\epsilon_t^2]$

Linear models



$$\pi_\beta(x) = \beta_0 + \beta_1 x = [\beta_0, \beta_1] \begin{bmatrix} 1 \\ x \end{bmatrix}$$

Two views of the problem

Learning as optimisation

- Each value $\pi_\beta(x)$ is a prediction about the value of y

Learning as inference

Two views of the problem

Learning as optimisation

- Each value $\pi_\beta(x)$ is a prediction about the value of y
- We suffer a loss $\ell(y, \pi_\beta(x))$ for every example (x, y) that we see.

Learning as inference

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Learning as optimisation

- Each value $\pi_\beta(x)$ is a prediction about the value of y
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- We can minimise the average loss over the data

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Learning as inference

- The parameters β define a probabilistic model $P_\beta(y|x)$ for every value of y.

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Learning as inference

- The parameters β define a probabilistic model $P_\beta(y|x)$ for every value of y.
- We want to find the parameters giving the highest probability on the observed data.

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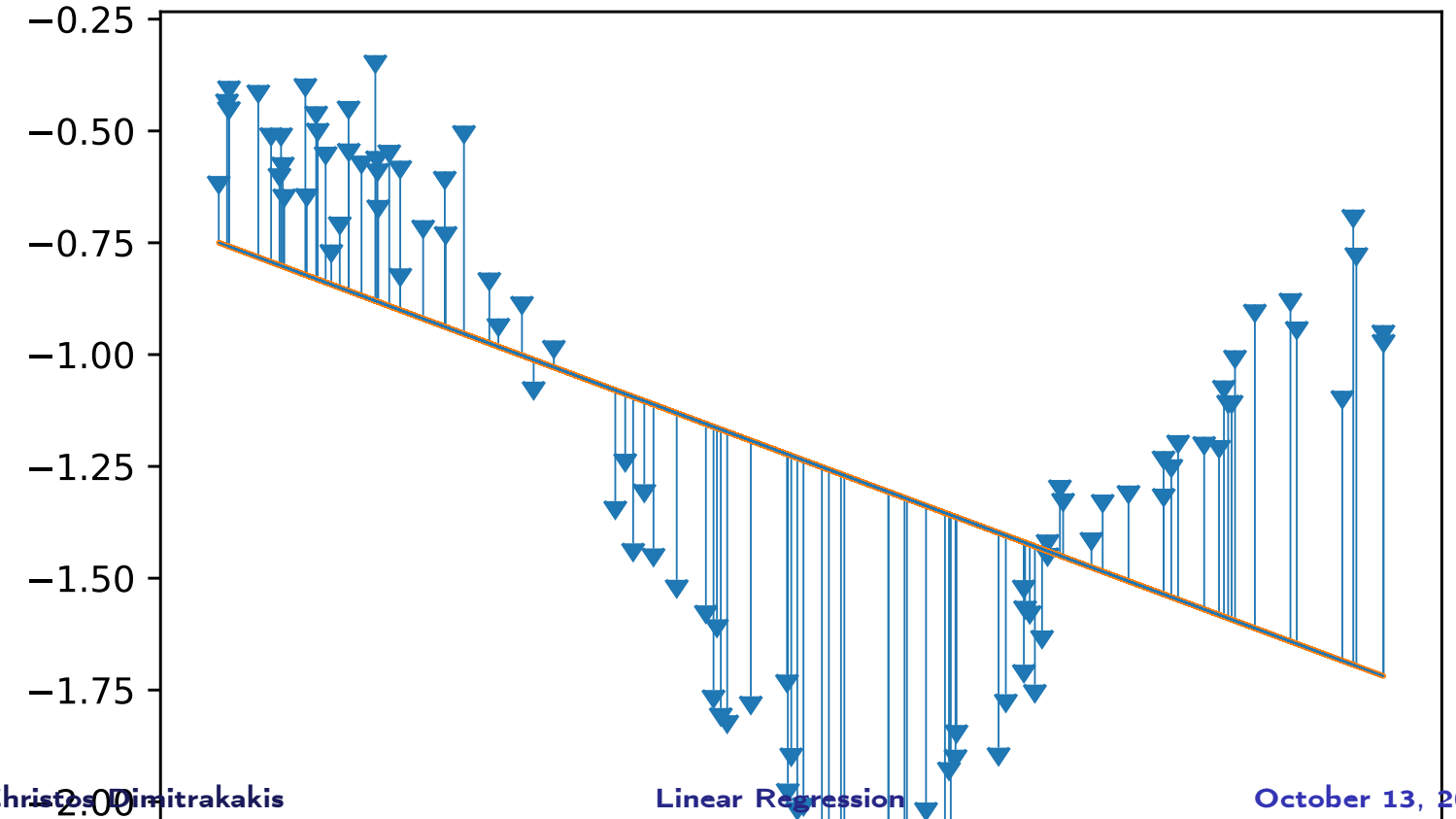
Learning as inference

- The parameters β define a probabilistic model $P_\beta(y|x)$ for every value of y.
- We want to find the parameters giving the highest probability on the observed data.
- Ideally, we want to find the true conditional distribution $P(y|x)$.

Learning as Optimisation

Find the parameters β minimising squared error

$$\min_{\beta} \frac{1}{T} \sum_{t=1}^{T} \Big[\underbrace{y^t - \pi_\beta(x^t)}_{\text{residual}} \Big]^2$$



Origins

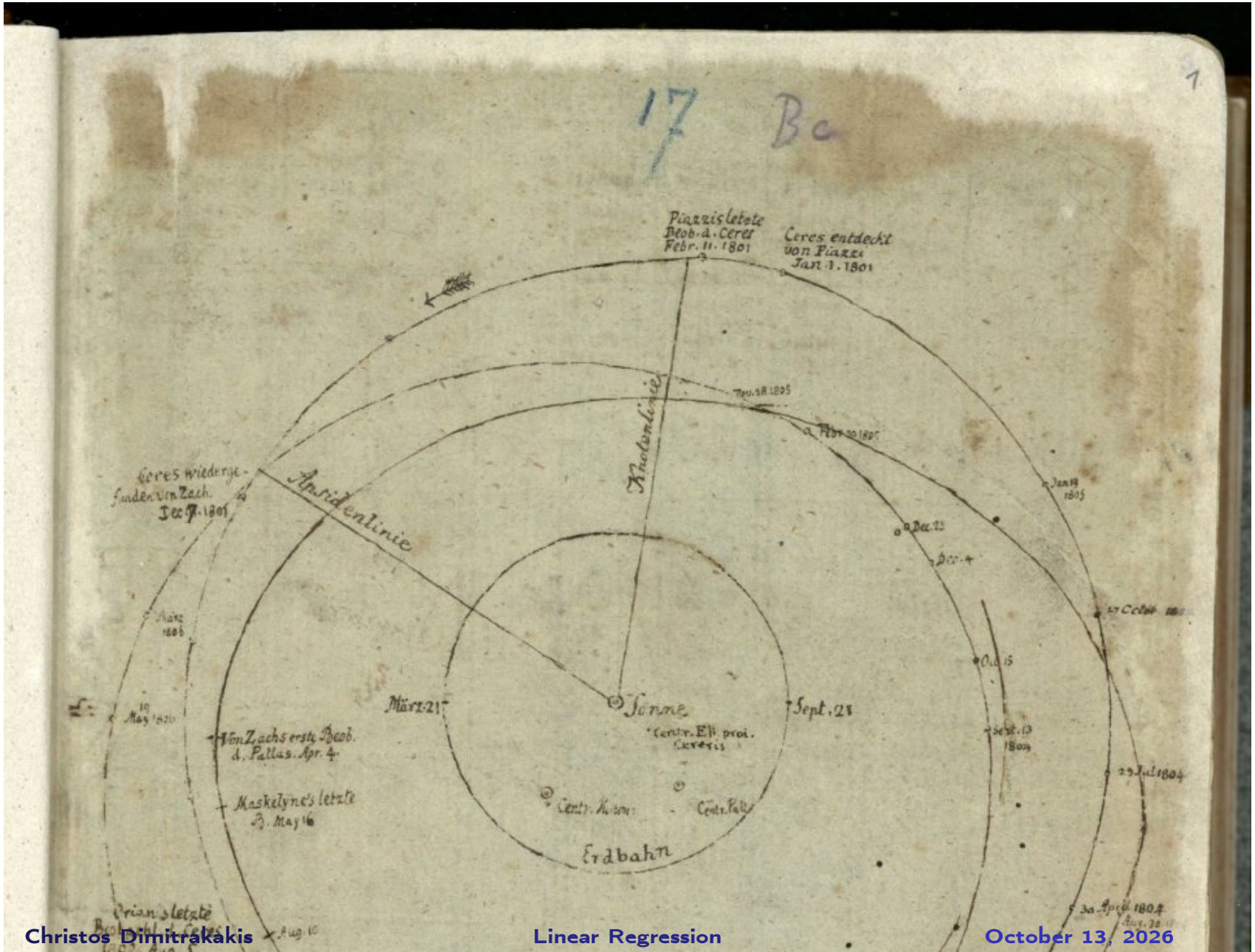


Figure: Gauss: originator



Figure: Legendre: first publication

The orbit of Ceres



Maximum likelihood inference

Gaussian noise model:

$$y^t = f(x^t) + \epsilon_t, \qquad \epsilon_t \sim \text{Normal}(0, \sigma)$$

With conditional density

$$p_\beta(y^t|x^t) \propto \exp(-[y^t - \pi_\beta(x^t)]^2/2\sigma^2)$$

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Idea: For data D, find parameters maximising $P_\beta(D)$

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$$\begin{aligned}
\arg\max_\beta P_\beta(D) &= \arg\max_\beta p_\beta(y^1, \ldots, y^t|x^1, \ldots, x^T) = \arg\max_\beta \ln \prod_t p_\beta(y^t|x^t) \\
&= \arg\max_\beta \sum \ln p_\beta(y^t|x^t) \\
&= \arg\max_\beta \sum_t \ln \left\{\exp\left(-[y^t - \pi_\beta(x^t)]^2/2\sigma^2\right)\right\} \\
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&= \arg\max_\beta \sum_t \ln \left\{ \exp\left(-[y^t - \pi_\beta(x^t)]^2/2\sigma^2\right) \right\} \\
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\end{aligned}$$

Coding break

src/Regression/RegressionDemonstration.ipynb

- Show implementation
- Fit and residuals
- Multiple draws from the distribution
- Fit on non-linear data?

Multiple linear regression

Input and output

- Data pairs (x^t, y^t), $t = 1, \ldots, T$.
- Input $x^t \in \mathbb{R}^n$
- Output $y^t \in \mathbb{R}^m$.

Multiple linear regression

Input and output

- Data pairs (x^t, y^t), $t = 1, \ldots, T$.
- Input $x^t \in \mathbb{R}^n$
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Point predictions: Modelling the conditional expectation $\mathbb{E}[y^t|x^t]$

- Parameters $\boldsymbol{B} \in \mathbb{R}^{n \times m}$
- Function $\pi_{\boldsymbol{B}} : \mathbb{R}^n \to \mathbb{R}^m$, defined as

$$\pi_\beta(x^t) = \boldsymbol{B}^\top x_t = \sum_{i=1}^{n} \beta_i x_{t,i}$$

Probabilistic predictions: Modelling the conditional probability $\mathbb{P}[y^t|x^t]$

- $y^t = \mathbb{E}[y^t|x^t] + \epsilon_t$, with $\epsilon_t \in \mathbb{R}^m$ being zero-mean noise.
- Noise covariance matrix $\Sigma = \mathbb{V}(\epsilon) = \mathbb{E}[\epsilon_t \mid \epsilon_t \top]$

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Gradient descent algorithm

Minimising a function

$$\min_{\beta} f(\beta) \leq f(\beta') \forall \beta', \qquad \beta^* = \arg\min_{\beta} f(\beta) \Rightarrow f(\beta^*) = \min_{\beta} f(\beta)$$

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Gradient descent for minimisation

- Input β_0
- For $n = 0, \ldots, N$:
- $\beta_{n+1} = \beta_n - \eta_n \nabla_\beta f(\beta_n)$

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Step-size η_n

- η_n fixed: for online learning
- $\eta_n = c/[c + n]$ for asymptotic convergence
- $\eta_n = \arg\min_\eta f(\beta_n + \eta \nabla_\beta)$: Line search.

Gradient descent for squared error

The cost function

$L(\beta, D) = \sum_{t=1}^{T}(y^t - \pi_\beta(x^t))^2 = \sum_{t=1}^{T} \epsilon_t^2$, with $\epsilon_t \triangleq y^t - \pi_\beta(x^t)$.

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Cost gradient

Using the chain rule of differentiation, $\nabla_\beta f(\epsilon) = \nabla_\epsilon f(\epsilon) \nabla_\beta \epsilon$.

$$\nabla_\beta L(\beta, D) = \nabla_\beta \sum_{t=1}^{T} \epsilon_t^2 = \sum_{t=1}^{T} \nabla_\beta \epsilon_t^2 = \sum_{t=1}^{T} \nabla_{\epsilon_t} \epsilon_t^2 \nabla_\beta \epsilon$$

$$= \sum_{t=1}^{T} 2\epsilon_t \nabla_\beta [y^t - \pi_\beta(x^t)] = \sum_{t=1}^{T} 2[y^t - \pi_\beta(x^t)][-\nabla_\beta \pi_\beta(x^t)]$$

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Parameter gradient for linear regression

Remember $\nabla_\beta f = (\partial/\partial_1 f, \ldots, \partial/\partial_n f)$

$$\frac{\partial}{\partial \beta_j} \pi_\beta(x^t) = \frac{\partial}{\partial \beta_j} \sum_{i=1}^{n} \beta_i x_{t,i} = \sum_{i=1}^{n} \frac{\partial}{\partial \beta_j} \beta_i x_{t,i} = x_{t,j}.$$

Stochastic gradient descent algorithm

When f is an expectation

$$f(\beta) = \int_X dP(x) g(x, \beta).$$

Replacing the expectation with a sample:

$$\begin{aligned} \nabla f(\beta) &= \int_X dP(x) \nabla g(x, \beta) \\ &\approx \frac{1}{K} \sum_{k=1}^{K} \nabla g(x^{(k)}, \beta), \qquad x^{(k)} \sim P. \end{aligned}$$

Analytical Least-Squares Solution

We need to solve the following equations for A:

$$\begin{aligned} y^1 &= (\boldsymbol{x}^1)^\top \boldsymbol{\beta} \\ \dots &\quad \dots \\ y^t &= (\boldsymbol{x}^t)^\top \boldsymbol{\beta} \\ \dots &\quad \dots \\ y^T &= (\boldsymbol{x}^T)^\top \boldsymbol{\beta} \end{aligned}$$

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We can rewrite it in matrix form:

$$\begin{pmatrix} y_1 \\ \vdots \\ y_t \\ \vdots \\ y_T \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_1^\top \\ \vdots \\ \boldsymbol{x}_t^\top \\ \vdots \\ \boldsymbol{x}_T^\top \end{pmatrix} \boldsymbol{\beta}$$

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Resulting in

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta}.$$

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$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta}.$$

How can we get $\boldsymbol{\beta}$?

Finding the β

We now have a linear equation,

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta}.$$

We want to solve for β. If $\boldsymbol{X}$ had an inverse $\boldsymbol{X}^{-1}$, we could obtain

$$\boldsymbol{X}^{-1}\boldsymbol{y} = \boldsymbol{X}^{-1}\boldsymbol{X}\boldsymbol{\beta} = I\boldsymbol{\beta} = \boldsymbol{\beta}.$$

But $\boldsymbol{X}^{-1}$ does not exist.

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Least-squares solution

The left-pseudo inverse $\tilde{\boldsymbol{X}}^{-1} \triangleq (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top$ can be used to obtain

$$\boldsymbol{\beta} = \tilde{\boldsymbol{X}}^{-1}\boldsymbol{y},$$

This follows as:

$$\begin{aligned}
\boldsymbol{y} &= \boldsymbol{X}\boldsymbol{\beta} \\
\tilde{\boldsymbol{X}}^{-1}\boldsymbol{y} &= \tilde{\boldsymbol{X}}^{-1}X\boldsymbol{\beta} \\
\tilde{\boldsymbol{X}}^{-1}\boldsymbol{y} &= \underbrace{(\boldsymbol{X}^\top \boldsymbol{X})^{-1}}_{\boldsymbol{A^{-1}}}\underbrace{\boldsymbol{X}^\top \boldsymbol{X}}_{\boldsymbol{A}}\boldsymbol{\beta}.
\end{aligned}$$

Some matrix algebra reminders

The identity matrix $\boldsymbol{I} \in \mathbb{R}^{n \times n}$

- For this matrix, $I_{i,i} = 1$ and $I_{i,j} = 0$ when $j \neq i$.
- $\boldsymbol{I}\boldsymbol{x} = \boldsymbol{x}$ and $\boldsymbol{I}\boldsymbol{A} = \boldsymbol{A}$.

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The inverse of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$

$\boldsymbol{A}^{-1}$ is called the inverse of $\boldsymbol{A}$ if

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The pseudo-inverse of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times m}$

- $\tilde{\boldsymbol{A}}^{-1}$ is called the left pseudoinverse of $\boldsymbol{A}$ if $\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A} = \boldsymbol{I}$.

$$\tilde{\boldsymbol{A}}^{-1} = (\boldsymbol{A}^\top \boldsymbol{A})^{-1}\boldsymbol{A}^\top, \qquad n > m$$

- $\tilde{\boldsymbol{A}}^{-1}$ is called the right pseudoinverse of $\boldsymbol{A}$ if $\boldsymbol{A}\tilde{\boldsymbol{A}}^{-1} = \boldsymbol{I}$.

$$\tilde{\boldsymbol{A}}^{-1} = \boldsymbol{A}^\top (\boldsymbol{A}\boldsymbol{A}^\top)^{-1}, \qquad m > n$$

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sklearn

Fitting a model to data

```
from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X, Y)
```

Getting predictions

We can get predictions for all inputs as an array

```
Z = model.predict(X)
```

Statsmodels

Fitting a model to data X, Y

```
import statsmodels.api as sm
Xa = sm.add_constant(X) # adds a constant factor to the data
model = sm.OLS(Y, Xa)
results = model.fit()
```

Getting predictions

The prediction is not just a point!

```
z = results.get_prediction(Xa[t])
z.predicted_mean # This is E[y|x]
```

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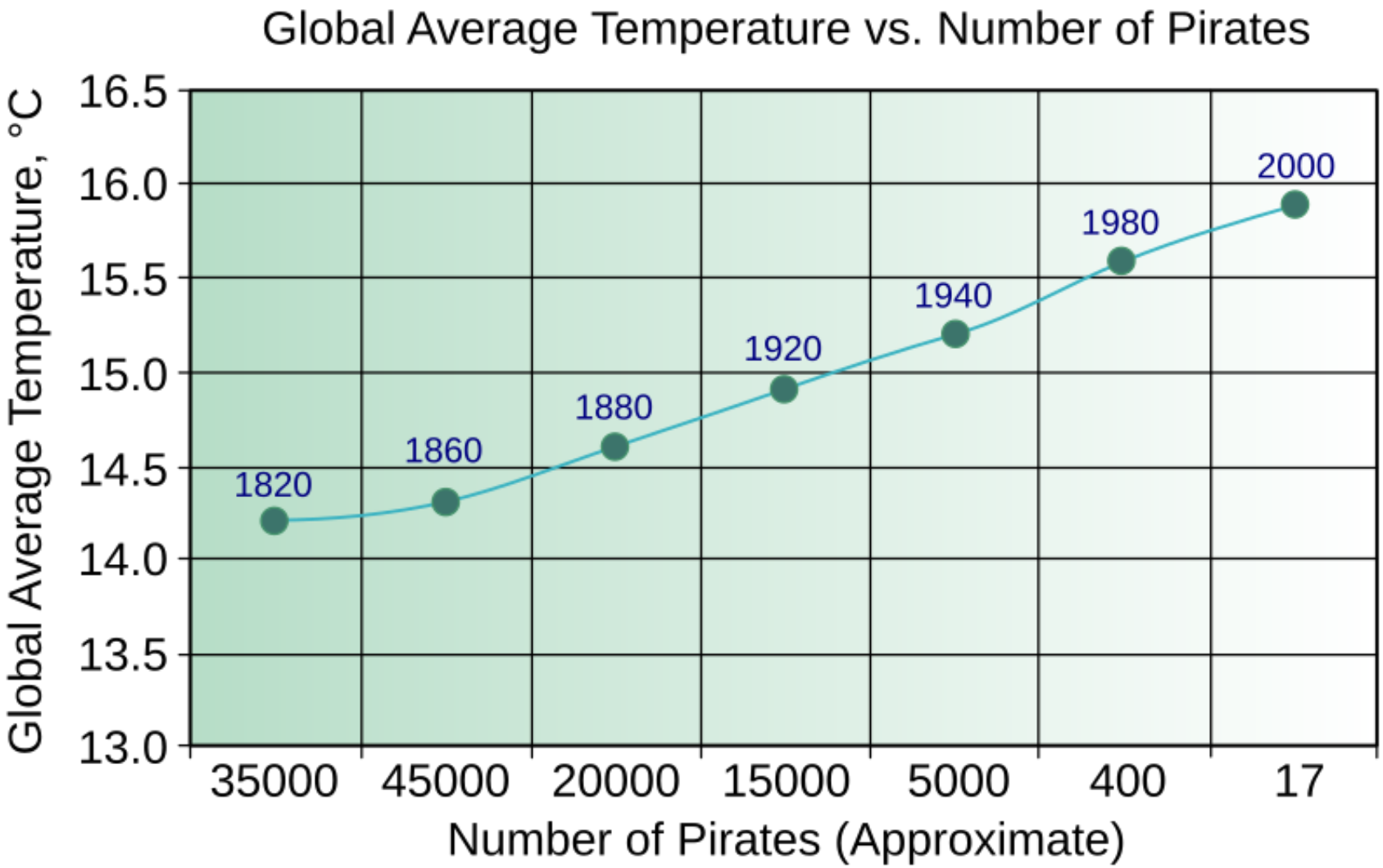
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Pitfalls

- β_i tells us how much y is correlated with $x_{t,i}$
- However, multiple correlations might be evident.
- Some features may be irrelevant
- The relationship may not be linear
- Correlation is not causation

Correlation is not causation



Linear regression exercises

- Exercises 8, 13 from ISLP
- A variant of Ex. 13 but with Y generated independently of X.